Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 100506

Subject: Power Electronics

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which device combines MOSFET input characteristics with BJT output characteristics?
(i) SCR
(ii) IGBT,
(iii) Diode
(iv) TRIAC

(b) Turn-on time for an SCR is 10 µsec. If an inserted in the anode circuit, then the turn-on time will be
(i) 10 µsec
(ii) more than 10 µsec
(iii) less than 10 µsec
(iv) about 10 µsec

(c) In a 3-phase semi-converter, for firing angle less than or equal to 60°, each thyristor and diode conduct, respectively, for
(i) 60°, 60°
(ii) 120°, 120°
(iii) 90°, 30°
(iv) 180°, 180°

(d) In a single-phase full converter, if output voltage has peak and average values of 325 V and 133 V respectively, then the firing angle is
(i) 40°
(ii) 50°
(iii) 140°
(iv) 130°

© Apna BEU

(e) A boost chopper is fed from a 220 V dc source to deliver a load voltage of 660 V. If the non-conduction time of the thyristor is 100 µs, the required pulse width would be
(i) 100 µs
(ii) 220 µs
(iii) 200 µs
(iv) 660 µs

(f) A chopper can be used on
(i) Pulse-width modulation Only
(ii) Amplitude modulation only
(iii) Frequency modulation only
(iv) Both PWM and FM

(g) A single-phase full bridge VSI has inductor L as the load. For a constant source voltage, the current through the inductor is
(i) Square wave.
(ii) Step function
(iii) Triangular wave
(iv) Pulsed wave

(h) A single-phase full bridge inverter can operate in load-commutation mode in case load consists of
(i) RL
(ii) RLC overdamped
(iii) RLC underdamped
(iv) RLC critically damped

(i) The power delivered to a star-connected load of R Ω per phase, from a 3-phase bridge inverter fed from fixed dc source, is 10 kW for 180° mode. From 120° mode, the power delivered to load would be
(i) 10 kW
(ii) 6.667 kW
(iii) 5 kW
(iv) 7.5 kW

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

(j) The output voltage waveform of a 3-phase square-wave inverter contains
(i) only even harmonics
(ii) only odd harmonics
(iii) both odd and even harmonics
(iv) only triplan harmonics

Q.2 (a) Explain the construction, working principle, and V-I characteristics of a SCR. [7]
(b) Explain voltage commutation with suitable circuit diagram and waveforms. [7]

Q.3 (a) Describe the operation and characteristics of a MOSFET used in power electronics applications. [7]
(b) Explain RC firing circuit for SCR with appropriate waveforms. [7]

Q.4 (a) Describe the working of a single-phase full converter in the inverter mode with RLE load. Illustrate your answer with waveforms for source voltage, E, load voltage and current, source current, current through and voltage across the SCR. Assume continuous conduction. [7]
(b) A resistive load of 10 Ω is connected through a half-wave SCR circuit to 220 V, 50 Hz, single-phase source. Calculate the power delivered to load for a firing angle of 60°. Find also the value of input power factor. [7]

© Apna BEU

Q.5 (a) A single-phase semi-converter feeds power to RLE load. For discontinuous load current, draw the source voltage, output voltage, load current, source current and freewheeling diode current waveforms as a function of time when [7]
(i) Extinction angle $\beta > \pi$
(ii) Extinction angle $\beta < \pi$ with $V_m \sin\beta < E$
(b) A 3-phase full converter is used for charging a battery with an emf of 110 V and an internal resistance of 0.2 Ω. For a constant charging current of 10 A, compute the firing angle delay for ac line voltage of 220 V. Find also the supply power factor. [7]

Q.6 (a) Explain the operation of a buck converter with circuit diagram and steady-state waveforms. Also derive the relationship between duty ratio and output voltage in a buck converter. [7]
(b) A chopper fed from a 220 V dc source, is working at a frequency of 50 Hz and is connected to an R-L load of R=5 Ω and L=40 mH. Determine the value of duty cycle at which the minimum load current will be (i) 5A (ii) 10 A (iii) 20 A and (iv) 30 A. [7]

Q.7 (a) Describe the principal of boost converter. Drive an expression for the average output voltage in terms of input voltage and duty cycle. State the assumption made. [7]
(b) Explain the operation of a single-phase voltage source inverter under square wave operation. [7]

Q.8 (a) Explain the operation of a three-phase voltage source inverter with switch states and output waveforms for 120o mode of conduction. [7]
(b) A single-phase full-bridge inverter delivers power to RLC load with R=3 Ω, X_L=12 Ω. The bridge operates with a periodicity of 0.2 ms. Calculate the value of C so that load commutation is achieved for the thyristors. Turn-off time for thyristors is 12μs. Factor of safety is 2. Assume the load current to contain only the fundamental component. [7]

Q.9 Write short note on *any two* of the following: [7 x 2 = 14]
(a) Gate driver circuit for MOSFET and IGBT
(b) Three-phase half wave rectifier with RLE load
(c) Time ratio control and frequency control in DC to DC Converter
(d) Pulse width modulation techniques

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play